

AIML PROJECT REPORT

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Course - Fundamentals of artificial intelligence and machine learning

Phishing Email Detector

1. Introduction

Phishing attacks are one of the most common cybersecurity threats where attackers attempt to trick users into revealing sensitive information such as passwords, bank details, or personal data. These attacks often come in the form of emails that appear legitimate.

This project aims to develop an AI-based Phishing Email Detector that can automatically classify emails as phishing or safe using Machine Learning techniques.

2. Problem Statement

Traditional phishing detection systems based on fixed rules or keyword matching are not reliable because:

- Attackers constantly change their language
- New phishing patterns emerge frequently
- Static rules fail to adapt to new threats

Therefore, a data-driven approach using Machine Learning is required to improve detection accuracy.

3. Objectives

- To detect phishing emails using Machine Learning
 - To improve accuracy over rule-based systems
 - To process and analyse real-world email data
 - To build a user-friendly detection system
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4. Methodology

4.1 Data Collection

The dataset was obtained from Kaggle and contains:

- Email text/content
- Labels indicating phishing (1) or safe (0)

4.2 Data Preprocessing

The dataset was cleaned and prepared by:

- Removing missing values
- Converting text to lowercase
- Preparing input (text) and output (labels)

4.3 Feature Extraction

Text data was converted into numerical form using:

- TF-IDF (Term Frequency - Inverse Document Frequency)
This technique assigns importance to words based on their frequency and relevance.

4.4 Model Training

The dataset was split into:

- Training set (80%)
- Testing set (20%)

A Logistic Regression model was used because:

- It is simple and efficient
- Works well for text classification problems

4.5 Model Evaluation

The model performance was evaluated using accuracy score, which measures how many predictions are correct.

4.6 Prediction System

A function was created to:

- Accept user input (email/message)
 - Convert it into vector form
 - Predict whether it is phishing or safe
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5. System Implementation

Tools & Technologies Used

- Programming Language: Python
 - Libraries:
 - pandas
 - scikit-learn
 - Development Environment: Visual Studio Code
 - Version Control: Git & GitHub
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6. Results

- The model successfully classifies emails as phishing or safe
 - Achieved good accuracy on test data
 - Performs better than keyword-based detection
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7. Conclusion

The project demonstrates how Machine Learning can be used to detect phishing emails effectively. By training on real-world data, the system provides more reliable and scalable detection compared to traditional methods.

8. References

- [Kaggle Phishing Email Dataset](#)
 - [Python Documentation](#)
 - [scikit-learn Documentation](#)
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9. Author

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